

LLM Fundamentals - Interview Ready Notes

What is LLM

LLM is a Transformer-based model that predicts the next token based on previous tokens. It learns patterns, not exact answers.

Tokenization

Breaks text into tokens and converts them to IDs. Controls cost and input size.

Embeddings

Convert text into vectors capturing meaning. Used in similarity search and RAG.

Vectors

Numerical representation of meaning in high-dimensional space.

Positional Embedding

Adds order information to tokens so sequence is understood.

Self-Attention

Allows tokens to focus on relevant tokens for context understanding.

Multi-Head Attention

Multiple attention heads capture different relationships.

Training

Model learns by predicting next token and updating weights using backpropagation.

Inference

Model generates output token-by-token using learned patterns.

Context Window

Maximum tokens model can process. Requires chunking in large data.

Temperature

Controls randomness of output.

System vs User Prompt

System defines behavior; user provides query.

Few-shot Prompting

Providing examples to guide model output.

Function Calling

LLM returns structured instruction to call backend function.

Tool Calling

LLM interacts with external tools like APIs.

Structured Output

LLM returns machine-readable formats like JSON.

LLM Limitation

LLM predicts, not retrieves, leading to hallucination.

Production Flow

User → Backend → Prompt → LLM → Tools/RAG → Structured Output → Response